

Masquerading

Sub-techniques (12)



Adversaries may attempt to manipulate features of their artifacts to make them appear legitimate or benign to users and/or security tools. Masquerading occurs when the name or location of an object, legitimate or malicious, is manipulated or abused for the sake of evading defenses and observation. This may include manipulating file metadata, tricking users into misidentifying the file type, and giving legitimate task or service names.

Renaming abusable system utilities to evade security monitoring is also a form of [Masquerading](#).
[\[1\]](#)

ID: T1036

Sub-techniques: [T1036.001](#), [T1036.002](#), [T1036.003](#), [T1036.004](#), [T1036.005](#), [T1036.006](#), [T1036.007](#), [T1036.008](#), [T1036.009](#), [T1036.010](#), [T1036.011](#), [T1036.012](#)

- ① **Tactic:** [Stealth](#)
- ① **Platforms:** Containers, ESXi, Linux, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
G1030	Agrius	Agrius used the Plink tool for tunneling and connections to remote machines, renaming it <code>systems.exe</code> in some instances. ^[2]
G1007	Aoqin Dragon	Aoqin Dragon has used fake icons including antivirus and external drives to disguise malicious payloads. ^[3]
S0622	AppleSeed	AppleSeed can disguise JavaScript files as PDFs. ^[4]
G0007	APT28	APT28 has renamed the WinRAR utility to avoid detection. ^[5]
G0050	APT32	APT32 has disguised a Cobalt Strike beacon as a Flash Installer. ^[6]
C0046	ArcaneDoor	ArcaneDoor involved the use of digital certificates on adversary-controlled network infrastructure that mimicked the formatting used by legitimate Cisco ASA appliances. ^[7]
S1246	BeaverTail	BeaverTail has masqueraded as MiroTalk installation packages: "MiroTalk.dmg" for macOS and "MiroTalk.msi" for Windows, and has included login GUIs with MiroTalk themes. ^[8]
S0268	Bisonal	Bisonal dropped a decoy payload with a .jpg extension that contained a malicious Visual Basic script. ^[9]
S0635	BoomBox	BoomBox has the ability to mask malicious data strings as PDF files. ^[10]
G0060	BRONZE BUTLER	BRONZE BUTLER has masked executables with document file icons including Word and Adobe PDF. ^[11]
C0015	C0015	During C0015 , the threat actors named a binary file <code>compareForfor.jpg</code> to disguise it as a JPG file. ^[12]
C0018	C0018	During C0018 , AvosLocker was disguised using the victim company name as the filename. ^[13]
G1052	Contagious Interview	Contagious Interview has delivered BeaverTail malware masquerading as legitimate software or applications. ^[14] ^{[15][16][17][8]} Contagious Interview has also delivered malicious payloads masquerading as legitimate software drivers. ^[18]
S0497	Dacls	The Dacls Mach-O binary has been disguised as a .nib file. ^[19]
S1111	DarkGate	DarkGate can masquerade as pirated media content for initial delivery to victims. ^[20]
S1066	DarkTortilla	DarkTortilla 's payload has been renamed <code>PowerShellInfo.exe</code> . ^[21]
S0673	DarkWatchman	DarkWatchman has used an icon mimicking a text file to mask a malicious executable. ^[22]
S9038	DynoWiper	DynoWiper has been named after well-known files schtask.exe, schtask2.exe, and _update.exe. ^{[23][24]}
G1003	Ember Bear	Ember Bear has renamed the legitimate Sysinternals tool procdump to alternative names such as <code>dump64.exe</code> to evade detection. ^[25]
S0634	EnvyScout	EnvyScout has used folder icons for malicious files to lure victims into opening them. ^[10]
G1016	FIN13	FIN13 has masqueraded staged data by using the Windows certutil utility to generate fake Base64 encoded certificates with the input file. ^{[26][27]}
S0696	Flagpro	Flagpro can download malicious files with a .tmp extension and append them with .exe prior to execution. ^[28]
S0661	FoggyWeb	FoggyWeb can masquerade the output of C2 commands as a fake, but legitimately formatted WebP file. ^[29]

ID	Name	Description
G0140	LazyScripter	LazyScripter has used several different security software icons to disguise executables. ^[33]
G0045	menuPass	menuPass has used esentuti to change file extensions to their true type that were masquerading as .txt files. ^[34]
S1015	Milan	Milan has used an executable named <code>companycatalogue</code> to appear benign. ^[35]
S0637	NativeZone	NativeZone has, upon execution, displayed a message box that appears to be related to a Ukrainian electronic document management system. ^[36]
G0133	Nomadic Octopus	Nomadic Octopus attempted to make Octopus appear as a Telegram Messenger with a Russian interface. ^[37]
S0368	NotPetya	NotPetya drops PsExec with the filename <code>dllhost.dat</code> . ^[38]
G0049	OilRig	OilRig has used .doc file extensions to mask malicious executables. ^[39]
C0016	Operation Dust Storm	For Operation Dust Storm , the threat actors disguised some executables as JPG files. ^[40]
C0006	Operation Honeybee	During Operation Honeybee , the threat actors modified the MaoCheng dropper so its icon appeared as a Word document. ^[41]
G0068	PLATINUM	PLATINUM has renamed <code>rar.exe</code> to avoid detection. ^[42]
S0453	Pony	Pony has used the Adobe Reader icon for the downloaded file to look more trustworthy. ^[43]
S1046	PowGoop	PowGoop has disguised a PowerShell script as a .dat file (<code>goopdate.dat</code>). ^[44]
S0565	Raindrop	Raindrop was built to include a modified version of 7-Zip source code (including associated export names) and Far Manager source code. ^{[45][46]}
S0458	Ramsay	Ramsay has masqueraded as a JPG image file. ^[47]
S0662	RCSession	RCSession has used a file named <code>English.rtf</code> to appear benign on victim hosts. ^{[48][49]}
S1240	RedLine Stealer	RedLine Stealer malware has masqueraded as legitimate software such as "PDF Converter Software" which has been distributed through poisoned search engine results often resembling legitimate software lures with the combination of typo squatted domains. ^[50]
S0148	RTM	RTM has been delivered as archived Windows executable files masquerading as PDF documents. ^[51]
S0446	Ryuk	Ryuk can create .dll files that actually contain a Rich Text File format document. ^[52]
S1018	Saint Bot	Saint Bot has renamed malicious binaries as <code>wallpaper.mp4</code> and <code>slideshow.mp4</code> to avoid detection. ^{[53][54]}
C0059	Salesforce Data Exfiltration	During Salesforce Data Exfiltration , threat actors used voice calls to socially engineer victims into authorizing a modified version of the Salesforce Data Loader app. ^[55]
G0034	Sandworm Team	Sandworm Team masqueraded malicious installers as Windows update packages to evade defense and entice users to execute binaries. ^[56]
S0615	SombRAT	SombRAT can use a legitimate process name to hide itself. ^[57]
G1046	Storm-1811	Storm-1811 has prompted users to download and execute batch scripts that masquerade as legitimate update files during initial access and social engineering operations. ^[58]

ID	Name	Description
S0682	TrailBlazer	TrailBlazer has used filenames that match the name of the compromised system in attempt to avoid detection. ^[62]
S0266	TrickBot	The TrickBot downloader has used an icon to appear as a Microsoft Word document. ^[63]
S1164	UPSTYLE	UPSTYLE has masqueraded filenames using examples such as <code>update.py</code> . ^[64]
S0689	WhisperGate	WhisperGate has been disguised as a JPG extension to avoid detection as a malicious PE file. ^[65]
G0112	Windshift	Windshift has used icons mimicking MS Office files to mask malicious executables. ^[66] Windshift has also attempted to hide executables by changing the file extension to ".scr" to mimic Windows screensavers. ^[67]
S0466	WindTail	WindTail has used icons mimicking MS Office files to mask payloads. ^[66]
G1035	Winter Vivern	Winter Vivern created specially-crafted documents mimicking legitimate government or similar documents during phishing campaigns. ^[68]
S0658	XCSSET	XCSSET installs malicious application bundles that mimic native macOS apps, such as Safari, by using the legitimate app's icon and customizing the <code>Info.plist</code> to match expected metadata. ^{[69][70]}
G0128	ZIRCONIUM	ZIRCONIUM has spoofed legitimate applications in phishing lures and changed file extensions to conceal installation of malware. ^{[71][72]}

Mitigations

ID	Mitigation	Description
M1049	Antivirus/Antimalware	Anti-virus can be used to automatically quarantine suspicious files.
M1047	Audit	Audit user accounts to ensure that each one has a defined purpose.
M1040	Behavior Prevention on Endpoint	Implement security controls on the endpoint, such as a Host Intrusion Prevention System (HIPS), to identify and prevent execution of potentially malicious files (such as those with mismatching file signatures).
M1045	Code Signing	Require signed binaries.
M1038	Execution Prevention	Use tools that restrict program execution via application control by attributes other than file name for common operating system utilities that are needed.
M1022	Restrict File and Directory Permissions	Use file system access controls to protect folders such as C:\Windows\System32.
M1018	User Account Management	Consider defining and enforcing a naming convention for user accounts to more easily spot generic account names that do not fit the typical schema.
M1017	User Training	Train users not to open email attachments or click unknown links (URLs). Such training fosters more secure habits within your organization and will limit many of the risks.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0127	Behavioral Detection of Masquerading Across Platforms via Metadata and Execution Discrepancy	AN0355	Adversary renames LOLBINs or deploys binaries with spoofed file names, internal PE metadata, or misleading icons to appear legitimate. File creation is followed by execution or service registration inconsistent with known usage.
		AN0356	Adversary drops renamed binaries in uncommon directories (e.g., /tmp, /dev/shm) or uses special characters in names (e.g., trailing space, Unicode RLO). Execution or cronjob registration follows shortly after file drop.
		AN0357	Adversary creates disguised launch daemons or apps with misleading names and bundle metadata (e.g., Info.plist values inconsistent with binary path or icon). Launch is correlated with user logon or persistence setup.
		AN0358	Adversary uses renamed container images, injects files into containers with misleading names or metadata (e.g., renamed system binaries), and executes them during startup or scheduled jobs.
		AN0359	Adversary places scripts or binaries with misleading names in /etc/rc.local.d or /var/spool/cron, or registers services with legitimate-sounding names not present in default ESXi builds.

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